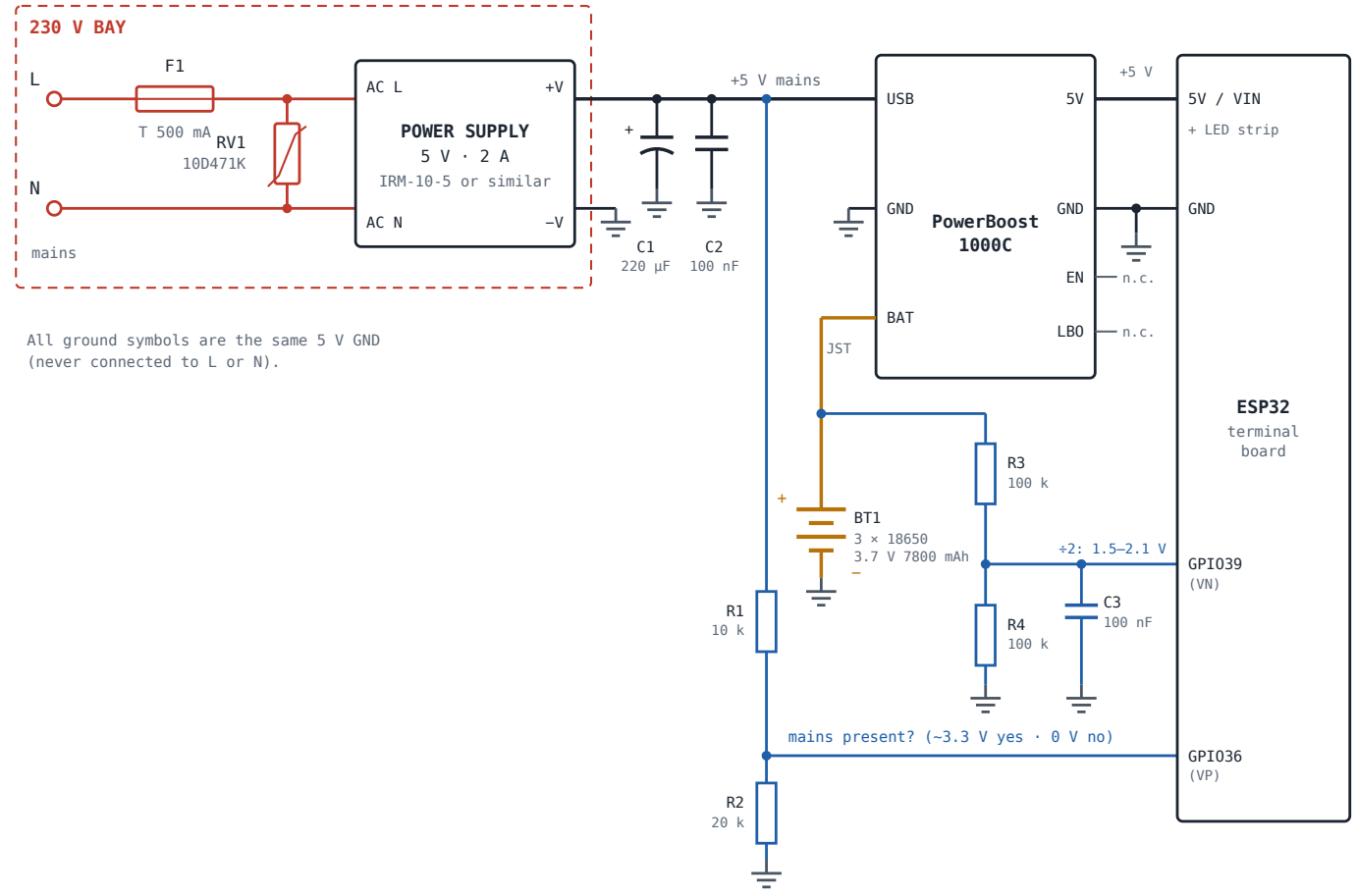
ESP32 NUKI FINGERPRINT ENTRANCE · BACKUP POWER

UPS wiring with PowerBoost 1000C

5 V 2 A power supply, PowerBoost 1000C, 3 × 18650 pack in parallel (3.7 V, 7800 mAh) and the ESP32 screw terminal board. At the bottom, the two dividers that tell the ESP32 whether mains power is present and how charged the battery is.



Connections

FROM	TO	NOTES
L mains	F1 → power supply AC L	fuse always on the live wire
N mains	power supply AC N	
RV1	between L and N	after the fuse
power supply +V	PowerBoost USB pad	C1 and C2 close to the power supply
power supply −V	PowerBoost GND pad	
+ battery	BAT (JST)	check the polarity with a multimeter
− battery	GND (JST)	
5V PowerBoost	ESP32 board 5V / VIN	the LED strip is fed from here too

Parts

REF.	VALUE	NOTES
F1	T 500 mA 250 V	5×20 slow-blow, with fuse holder
RV1	10D471K	470 V varistor
—	5 V 2 A	Mean Well IRM-10-5 or equivalent
C1	220 µF 16 V	electrolytic, stripe = −
C2, C3	100 nF 50 V	ceramic
U1	PowerBoost 1000C	Adafruit, without the USB-A socket
BT1	3 × INR18650L26	3 cells in parallel (3.7 V nominal)
R1	10 kΩ	brown black orange
R2	20 kΩ	red black orange (or 22 k)
R3, R4	100 kΩ	brown black yellow

FROM	TO	NOTES
`GND` PowerBoost	ESP32 board GND	
power supply +V	R1 → GPIO36 → R2 → GND	"mains present" sensor
`BAT`	R3 → GPIO39 → R4 → GND	C3 between GPIO39 and GND
`EN`, `LBO`	not connected	LBO must not go to a GPIO

Before powering up

1. Connect the **battery** to the PowerBoost first and measure `5V`–`GND` at the output: you should read about 5.1–5.2 V. The PowerBoost does not work without a battery.
2. With the ESP32 disconnected, power the supply and measure GPIO36 (on the wire that will go to the pin): about 3.3 V with mains, 0 V without.
3. Measure the point between R3 and R4: it must be half the battery voltage, i.e. 2.0 V if the pack is at 4.0 V.
4. Only then connect the ESP32 board and try cutting the mains: the ESP32 must not reboot.

> ### Safety
>
> - The battery pack and the PowerBoost sit **outside the 230 V bay**, away from the power supply, which gets warm.
> - The PowerBoost charger chip gets very hot while charging: that's normal, but keep it somewhere ventilated.
> - No battery wire may be left bare: a shorted 18650 heats up within seconds.
> - R1–R2 take the 5 V from the power supply output, never from the 230 V side.